



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to VANGALAMMAN HIGHWAY SERVICE

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

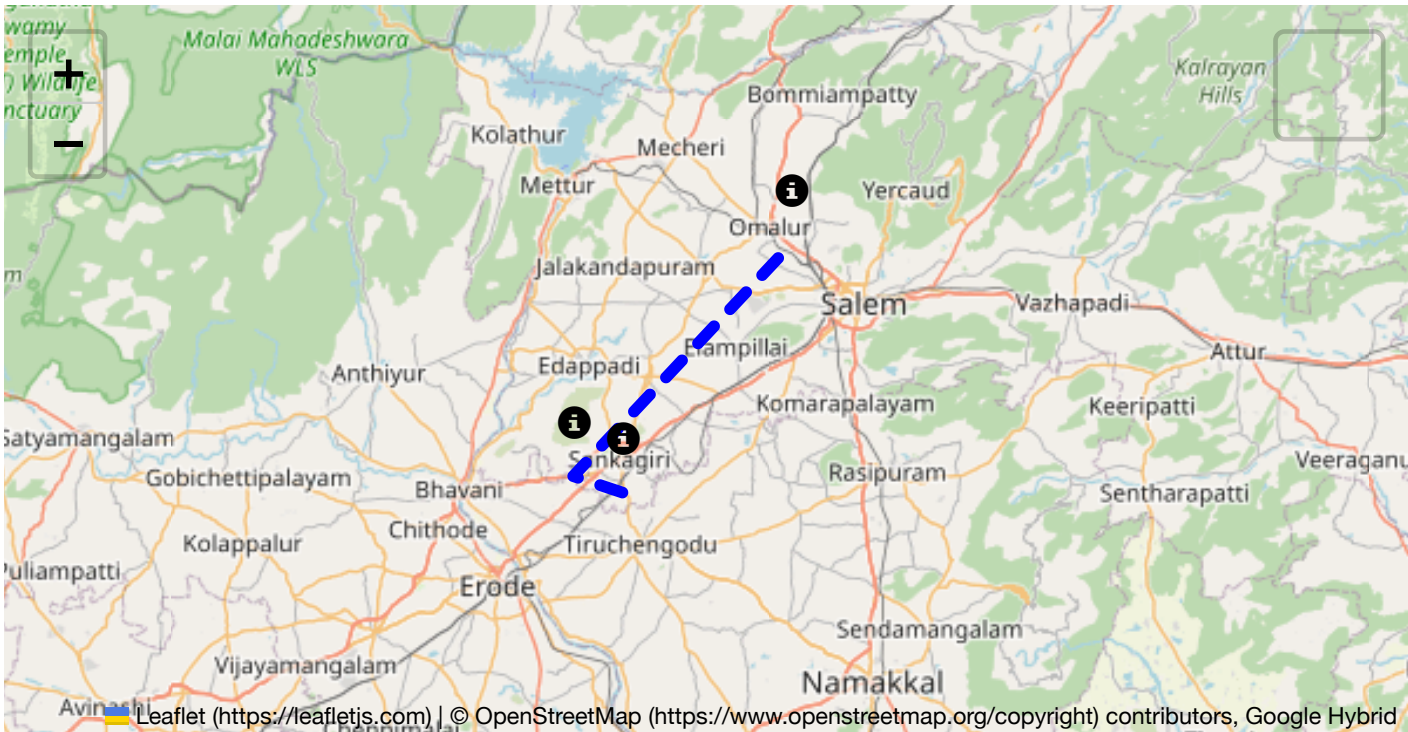
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.

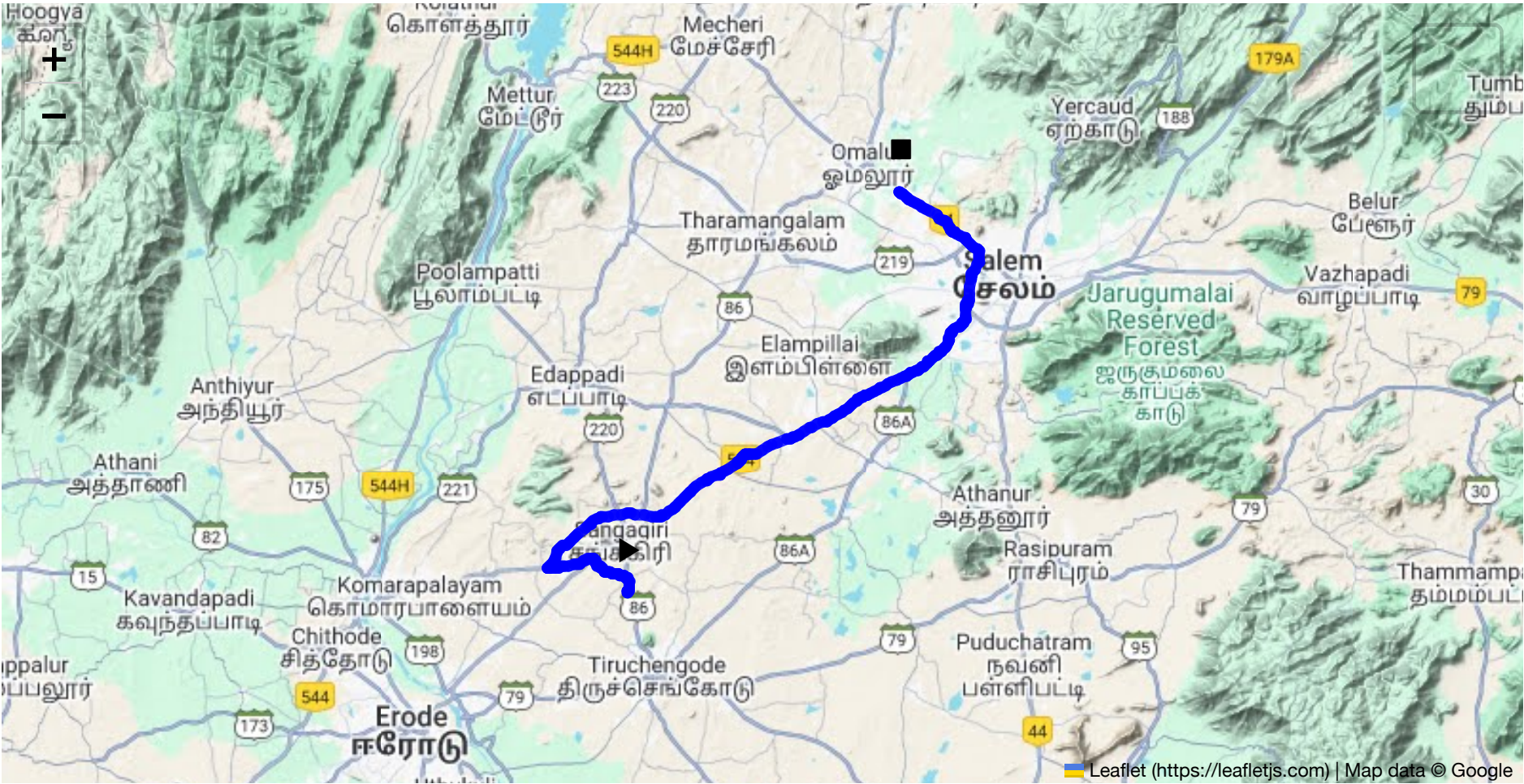


Route Summary:
Total Distance: 62.06 km
Estimated Duration: 1.2 hours

Adjusted Duration (Heavy Vehicle): 1.5 hours

Start: (11.4381, 77.8734)

End: (11.722192, 78.070355)



Welcome to the Journey Risk Management Study

To provide a comprehensive safety analysis for the specified route from Sangagiri to Kottagoundampatti via Padaiveedu, here is a detailed report covering various aspects that are crucial for a truck driver transporting hazardous materials.

1. Overview of the Route Map

The route spans approximately 62.06 kilometers, primarily on state highways and major district roads. It begins at Sangagiri, passing through Padaiveedu, and continues to Thannithotty, Salem. The route involves traversing mainly rural areas with occasional urban stretches as you approach Salem.

2. Typical Weather Conditions and Hazards

The region experiences a tropical climate with high temperatures, especially in summer (March to June). Monsoon seasons (June-September and October-December) can bring heavy rains, leading to slippery roads and potential flooding, particularly in low-lying areas. Fog is less common but possible during winter mornings and could impede visibility.

3. Traffic Patterns and Peak Hours

Traffic is generally light to moderate except in Salem during peak hours (8:00-10:00 AM and 5:00-7:00 PM). Areas near market hubs and school zones in small towns may see brief congestion. Expect slower movement in residential and commercial areas, especially as you near city limits.

4. Road Quality and Infrastructure

The road conditions are mixed. State highways are relatively well-maintained with proper signage and lane marking, while smaller district roads may have potholes or uneven surfaces. Adequate lighting is

present in urban areas but less so in rural stretches. Road widening activities or maintenance work can periodically cause delays.

5. Alternative Routes for Emergencies

In case of emergencies or road closures, an alternative route is via NH44, connecting Sangagiri to Salem directly. However, this would be a longer detour and may require permission for carrying hazardous materials. Local routes through Omalur might also offer relief but verify accessibility and road conditions in advance.

6. Local Regulations on Hazardous Material Transport

Tamil Nadu imposes strict regulations on the transport of hazardous materials, including restrictions on time of travel and designated routes. Transporters must carry appropriate documentation, adhere to permissible load limits, and ensure their vehicles have necessary placards and safety measures.

7. Historical Incidents Involving Heavy Vehicles

This route has seen a few incidents involving heavy vehicles, often due to brake failures on inclines or sharp turns, typically in the Salem outskirts. Fatal incidents are rare but highlight the need for cautious driving and regular vehicle maintenance.

8. Environmental Considerations and Sensitive Areas

The route passes close to sensitive ecological zones and agricultural lands; drivers should be wary of local wildlife crossings, especially near forest patches. Spillage or accidents involving hazardous materials could significantly impact these environments.

9. Communication Coverage

Most of the route offers adequate mobile network coverage, though certain rural stretches approaching Padaiveedu may experience brief dead zones. It's advisable to equip vehicles with GPS and emergency communication devices.

10. Estimated Emergency Response Times

Emergency response times vary:

- Urban areas near Salem generally see responses within 20-30 minutes.
- Rural sections may experience delays up to an hour due to distance and road accessibility. Ensure vehicle operators have access to emergency contacts and basic first aid.

11. Overall Summary of Risk Assessment

This route is fairly safe for transporting hazardous materials when observing all safety guidelines. The main risks involve weather conditions during monsoons, traffic congestion near Salem, and potential road maintenance issues. Keeping abreast of weather updates, traffic reports, and adhering to local regulations can mitigate these risks effectively. Regular vehicle checks and awareness of alternative paths ensure preparedness for varied scenarios.

Conclusion

This analysis should assist in preparing truck drivers for safe transit. Emphasizing cautious driving, especially during adverse weather, and maintaining communication ensures the safety of both the driver and the surrounding environment.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	Medium	11.43822, 77.87348	30 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	Medium	11.45358, 77.85781	30 KM/Hr
4	Turn	Medium	11.45353, 77.85697	30 KM/Hr
5	Turn	Medium	11.45839, 77.85143	30 KM/Hr
6	Turn	Medium	11.46053, 77.85034	30 KM/Hr
7	Turn	Medium	11.46307, 77.84941	30 KM/Hr
8	Turn	Medium	11.46316, 77.84921	30 KM/Hr
9	Turn	High	11.45542, 77.81725	15 KM/Hr
10	Blind Spot	Blind Spot	11.45631, 77.81668	10 KM/Hr
11	Turn	Medium	11.64183, 78.11810	30 KM/Hr
12	Turn	Medium	11.68920, 78.11799	30 KM/Hr
13	Turn	Medium	11.69386, 78.10968	30 KM/Hr
14	Turn	Medium	11.69392, 78.10964	30 KM/Hr
15	Turn	High	11.69400, 78.10963	15 KM/Hr
16	Turn	High	11.72224, 78.07065	15 KM/Hr
17	Turn	High	11.72218, 78.07057	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
4	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
5	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
6	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
9	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
13	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
14	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
15	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
16	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
17	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
18	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
20	hospital	Sri Gokulam Speciality Hospital	11.6788724, 78.1274258	30 km/h	Medium
21	hospital	Global Medical Center	11.6867049, 78.1186697	30 km/h	Medium
22	hospital	Manipal Hospital	11.703072, 78.1026612	30 km/h	Medium
25	pharmacy	MJ Medicals	11.7232728, 78.0698965	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
8	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
10	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
11	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
12	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
19	college	Sona College	11.6768, 78.1257052	30 km/h	Medium
23	college	Govt Engineering College	11.7128537, 78.0877084	30 km/h	Medium
24	university	Periyar University	11.7186758, 78.0771089	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45358, 77.85781



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45839, 77.85143



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46053, 77.85034



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46316, 77.84921



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.45631, 77.81668



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68920, 78.11799



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.69386, 78.10968



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.69392, 78.10964



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.69400, 78.10963



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.72224, 78.07065



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.72218, 78.07057
